Reconstructing albedo

from mean cloud properties

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MOTIVATION

Kernel-based decompositions of cloud albedo are widely used in radiative forcing calculations. We assess **how well mean cloud properties (CF, LWP, N_d) reconstruct the albedo (α) of liquid cloud scenes** across the globe.

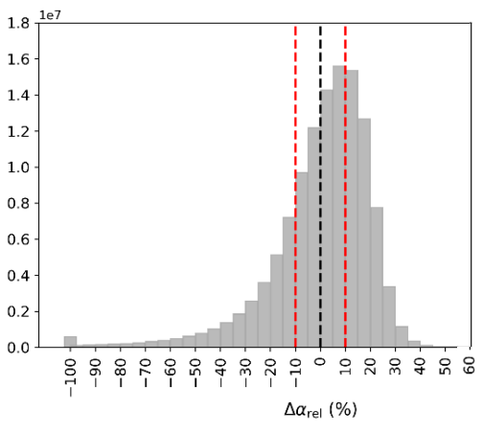
DATA & METHODS

- Data: **MODIS** and **CERES** retrievals, 60°S–60°N, 2003–2021.
- Method: a **3D joint histogram (kernel) of CF, LWP and N_d**, where bin-averaged albedo is compared to observed CERES albedo at each datapoint.

TAKEAWAY MESSAGE

Although **geographical patterns exist** in the CF–LWP–N_d–α relationship, **significant uncertainties remain**. Regional analysis reduces reconstructed biases, suggesting explanatory variables beyond mean CF, LWP and N_d.

KEY FINDINGS

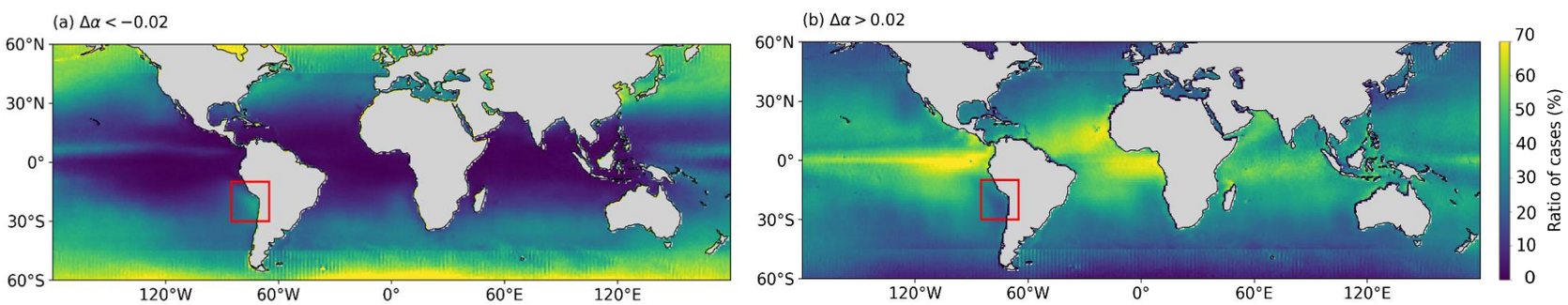


▲ ***Fig. 1.*** *Distribution of $\Delta\alpha_{rel}$ between bin-averaged and observed CERES albedo. Black dashed line: zero difference; red dashed lines: ±10% accuracy thresholds.*

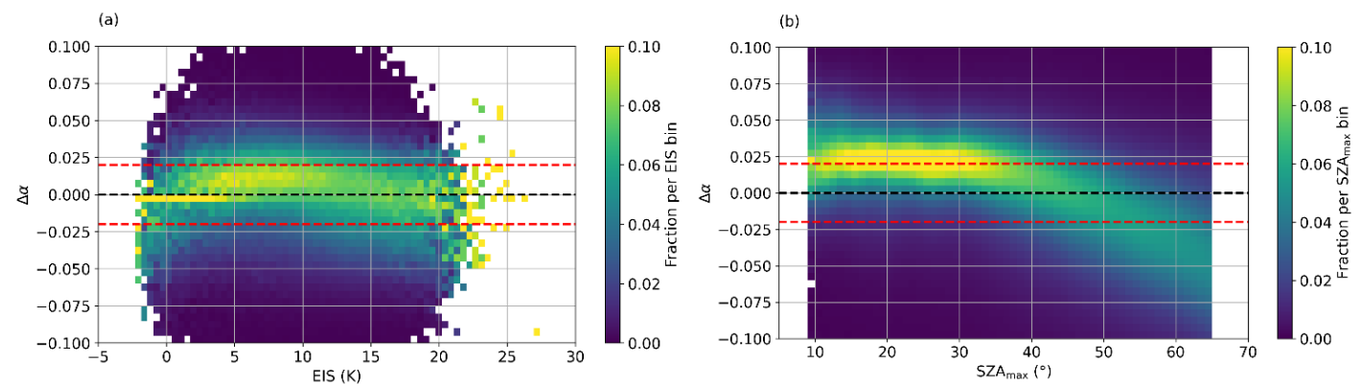
Reconstructed albedo **biases exceed ±10%** in up to **~60% of datapoints** (Fig. 1).

Albedo is **underestimated in marine stratocumulus** regions and **overestimated in the tropics** (Fig. 2).

In stratocumulus regions, biases follow an EIS-dependent pattern: thick stratocumulus near the coastline (high EIS) is underestimated, while shallow cumulus further offshore (low EIS) is overestimated (Fig. 3a).



▼ ***Fig. 2.*** *Percentage of cases with (a) underestimated and (b) overestimated albedo. Grey: land. Red rectangle: marine stratocumulus region used in Fig. 3a.*



▲ ***Fig. 3.*** *Joint histograms of (a) EIS–Δα for the 10°S–30°S, 85°W–65°W region (Aqua, 2007–2019) and (b) SZAmax–Δα for the global ocean (2003–2021). Columns normalised to show conditional probabilities. Red dashed lines: Δα = ±0.02, black dashed line: Δα = 0.00.*

Improving albedo estimates

- Accounting for **solar zenith angle** (Fig. 3b) improves estimates by 12.3–17.6%. The improvement is greater for overestimates than for underestimates.
- Calculating the kernel on a **higher spatial resolution** grid significantly improves estimates, although evaluating bin sparsity is necessary.
- Including **thin cloud fraction** in the kernel alongside CF, LWP and N_d only improves estimates by 0.9–1.6%.
- Constructing the kernel from **monthly-averaged** CF, LWP and N_d improves estimates by 16.8%.

Details: Wojciechowska, I. and Gryspeerdt, E. (2025). Reconstructing albedo from mean cloud properties, EGUsphere [preprint], https://doi.org/10.5194/egusphere-2025-4784.

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